

MA 301
Final
Fall - 2011

Problem	points	out of
1		10
2		10
3		10
4		10
5		10
6		10
7		10
Score		70

Name: _____

HR: _____

Signature: _____

READ BEFORE YOU START
* Show all work. * Change your calculator mode to RADIANS. * Sloppy solutions may be penalized. * Use calculators ONLY for computations. * You must write up every problem in a clear and organized way. * If you write solutions on a page different than the question, then clearly state where it is located.

Problem 1: Given $\mathbf{F} = [2, 2y + z, y + 1]$

(a) Show that $\mathbf{F}$ is conservative in two different ways

method 1

method 2

(b) Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ when C is a straight line from $(-1, 0, 2)$ to $(0, 2, -6)$

Problem 2: Let $\mathbf{F} = [y, x^2 \sin y]$ and C be the boundary of the region bounded by $y = \frac{\pi}{2}x$, $y = \frac{\pi}{2}$ and $0 \leq x \leq 1$ in the x, y -plane.

Sketch C and compute $\oint_C \mathbf{F} \cdot d\mathbf{r}$ using green's theorem.

Problem 3: Let $\mathbf{F} = [e^x, e^y, e^z]$ and S be the surface of the box $|x| \leq 1, |y| \leq 2, 0 \leq z \leq 1$.

Compute $\iint_S \mathbf{F} \cdot \mathbf{n} \, dA$

Problem 4: Answer the following questions

(a) If the integral $\oint_C \mathbf{F} \cdot d\mathbf{r}$ represents the work done by a object then explain the physical role of the vector-valued function $\mathbf{F}$ and the curve C .

(b) Suppose $f(x) = f(x + 6)$ for all x , then we say f is periodic with period 3. TRUE or FALSE

(c) $\sqrt{-5} \cdot \sqrt{-5} = 5$ TRUE or FALSE

(d) Let $\mathbf{F}$ vector field such that $\text{curl } \mathbf{F} = \mathbf{0}$. Match each expression in the left column with the most appropriate term in the right column.

$$\int_C \mathbf{F} \cdot d\mathbf{r}$$

conservative

$$\mathbf{F} \cdot d\mathbf{r}$$

zero

$$\oint_C \mathbf{F} \cdot d\mathbf{r}$$

path independent

$$\mathbf{F}$$

exact

(e) Match each the Theorem in the left column with the most appropriate statement in the right column.

Divergence Theorem

The surface of a sphere

Stokes' Theorem

Top-half of the surface of a sphere

Green's Theorem

The surface of a circle

Problem 5: Let $f(x) = \begin{cases} 0 & -2 < x < -1 \\ 1+x & -1 < x < 0 \\ 1-x & 0 < x < 1 \\ 0 & 1 < x < 2 \end{cases}$ and $f(x) = f(x+4)$

(a) Graph $f(x)$ on the interval $[-3, 5]$

(b) Find the Fourier Series of $f(x)$

Problem 6: Let $f(x) = \begin{cases} -x^2 & -1 < x < 0 \\ x^2 & 0 < x < 1 \end{cases}$ and $f(x) = f(x + 2)$

(a) Graph $f(x)$ on the interval $[-3, 3]$

(b) Can we say that any of Fourier coefficients a_0 , a_n , or b_n are zero? Why or why not?

(c) **Set-up**, but **DO NOT** compute the integral(s) for all the non-zero Fourier coefficients.

Problem 7: Let $z_1 = -2 - 2i$, $z_2 = i$. Answer the following:

(a) Compute the operations and write the result in standard form

(i) $z_1 z_2 =$

(ii) $z_1 - z_1 z_2 =$

(iii) $|z_1 z_2| =$

(iv) $(z_2)^{76}$

(d) Solve for z : $z_2 z^{1/3} = z_1$